

COSC 2793 Assignment 3 Report

1. Introduction & Task Definition

1.1 Project Overview

In modern clinical oncology, histopathology the microscopic structural examination of biological tissue sections serves as the gold standard for diagnosing complex conditions such as colorectal cancer. Manual examination of these tissue slices under a microscope is a highly demanding, labor-intensive, and time-critical process that is inherently susceptible to inter-observer variability among pathologists.

Automating the detection and categorization of individual cellular nuclei using digital pathology images offers scalable support tools for clinical workflows, enabling rapid, consistent, and objective multi-patient diagnostic sweeps. This project designs, implements, and evaluates an end-to-end computational machine learning pipeline using a modified version of the benchmark CRCHistoPhenotypes dataset (Sirinukunwattana et al., 2016) to classify individual cell profiles captured within restricted 27 x 27 pixel RGB image arrays.

1.2 Problem Statement

For this project, the task is split into two distinct supervised learning classification pipelines based on input features and biological attributes associated with the patient properties:

1. **Task 1 — Binary Classification (isCancerous):** Predict whether an input cell image represents a malignant (cancerous) structure or a benign/healthy cell. This task merges data from `data_labels_mainData.csv` and `data_labels_extraData.csv`, operating across 20,280 images collected from 98 unique patients.
2. **Task 2 — Multiclass Classification (cellType):** Categorize an input cell image into one of four mutually exclusive biological cell types: Fibroblast, Inflammatory, Epithelial, or Others. This task relies solely on `data_labels_mainData.csv`, tracking 9,896 individual cell slices across 60 patients.

1.3 Learning Objectives

The objectives of this project are to design and implement a complex deep learning and machine learning pipeline, evaluating Logistic Regression, Support Vector Machines (SVM), Random Forests, and custom Deep Convolutional Neural Networks (CNN). By evaluating these contrasting algorithms, this project demonstrates data preprocessing techniques, class imbalance handling, hyperparameter tuning, model evaluation, and a deep, multi-task understanding of the bias-variance trade-off linked to real-world clinical data distributions.

1.4 Performance Measures

To evaluate these models, there will be three primary evaluation metrics applied uniformly to test results:

- **Accuracy:** Captures the global percentage of correct predictions. This metric introduces significant bias under class imbalance, as a naive predictor can score highly by simply selecting the majority class.
- **Macro F1-score:** The primary metric for model tuning and selection. It calculates the F1-score for each class independently and averages them with equal weight, ensuring that minority populations carry equal predictive importance.
- **Confusion Matrices:** Map explicit true-versus-predicted layouts to identify exactly which cell profiles are hard to differentiate at a low image resolution.

For the binary isCancerous task, Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) statistics establish threshold-independent discriminative capacity. For deep architectures, training and validation loss trajectories across epochs diagnose underfitting, overfitting, and structural optimization stability.

2. Data Description & Exploratory Data Analysis

This section details the critical initial steps of understanding the dataset before any predictive modeling. The exploration focuses on the structure, relationships, and distributions within the histopathology data, identifying potential challenges that inform subsequent preprocessing and model selection.

2.1 Dataset Overview

The primary data spans a total profile of 20,280 samples across 98 unique patients with completely clean tracking. An initial review reveals that no data is missing from any instance or column, meaning that no data replacement or imputation strategies need to be implemented. Every raw cell sample maps to an individual 27 x 27 RGB image grid where pixel values span a dynamic intensity scale from a minimum of 78 to a maximum of 255. A metadata overview tracking structural rows, column configurations, shapes, and counts can be found in Table 1 below.

Dataset Target Group	Columns Contained	Row Entries Count	Feature Extraction Layout	Missing Values
Main Dataset	InstanceID, patientID, ImageName, cellTypeName, cellType,	9,896 Slices	27 x 27 x 3 Tensor (RGB)	None

	isCancerous			
Extra Dataset	InstanceID, patientID, ImageName, isCancerous	10,384 Slices	27 x 27 x 3 Tensor (RGB)	None

Table 1: Metadata overview of the feature structures within the histopathology classification partitions.

To understand the nature of the target variables, exploratory distributions were generated to visualize frequency across the patient metrics. The target variable breakdown can be found in Figure 1 below.

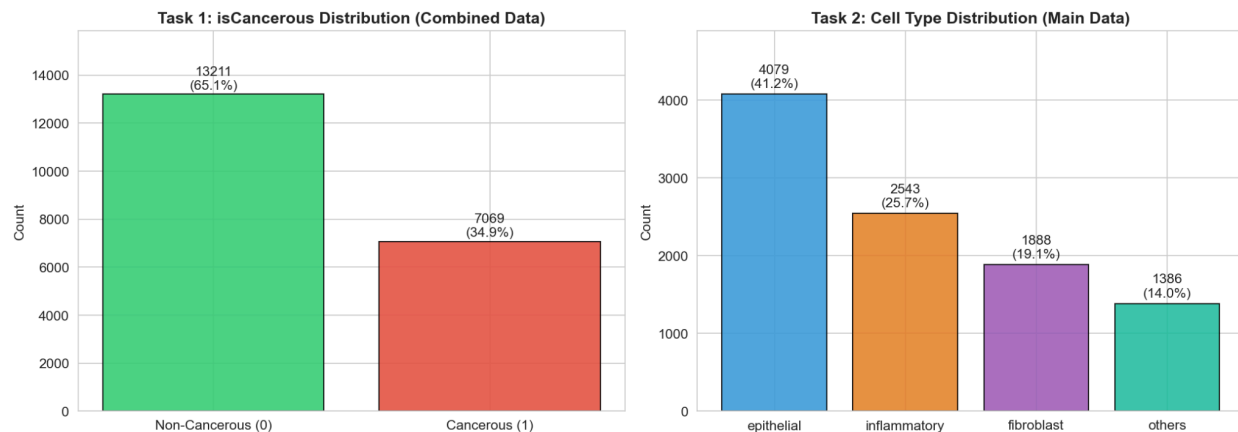


Figure 1: Target variable distributions for both tasks.

The class metrics reveal a distinct data skewing across both classification tasks:

- **isCancerous (Combined Task 1):** The dataset consists of 13,211 Non-Cancerous samples (65.1%) and 7,069 Cancerous samples (34.9%), establishing a moderate baseline imbalance.
- **cellType (Main Task 2):** The multi-class distribution is heavily skewed toward Epithelial structures with 4,079 samples (41.2%), followed by Inflammatory with 2,543 (25.7%), Fibroblast with 1,888 (19.1%), and Others with 1,386 (14.0%).

2.2 Statistical Summary

Understanding the variations and spreads of sample metrics per unique group is essential. Grouping the instances by patientID highlights significant variation in sample contributions. Task 1 shows patient sample frequencies ranging from a minimum of 6 images to a maximum of 699 images per patient ($\mu = 206.9$). Task 2 tracks a patient sample span between 11 and 389 images ($\mu = 164.9$). The analytical data leads to a right-skewed patient sample distribution,

which can be confirmed visually with Figure 2 below.

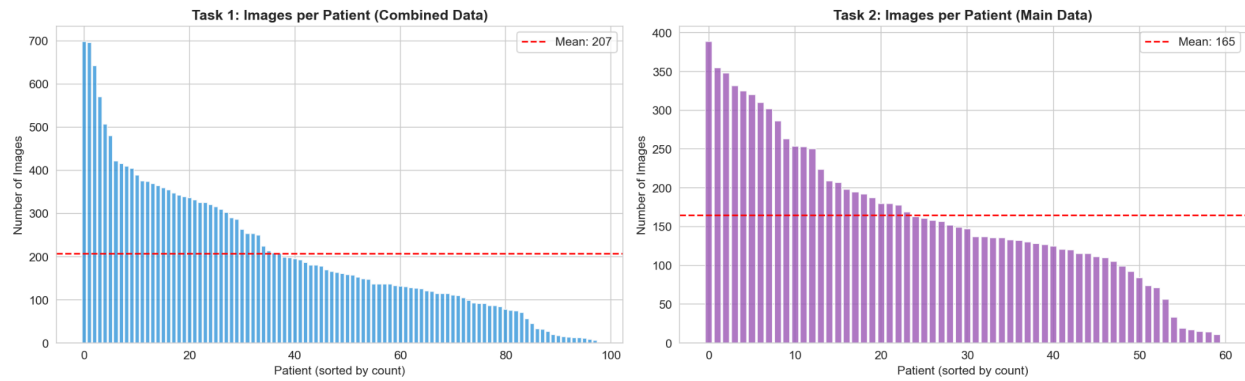


Figure 2: Patient distribution metrics

2.3 Visual Analysis & Color Channel Saturation

To extract deep structural features, individual image vectors were compiled to isolate average pixel channel illumination trends across the classification targets.

Task 1: Sample Images — Cancerous vs Non-Cancerous

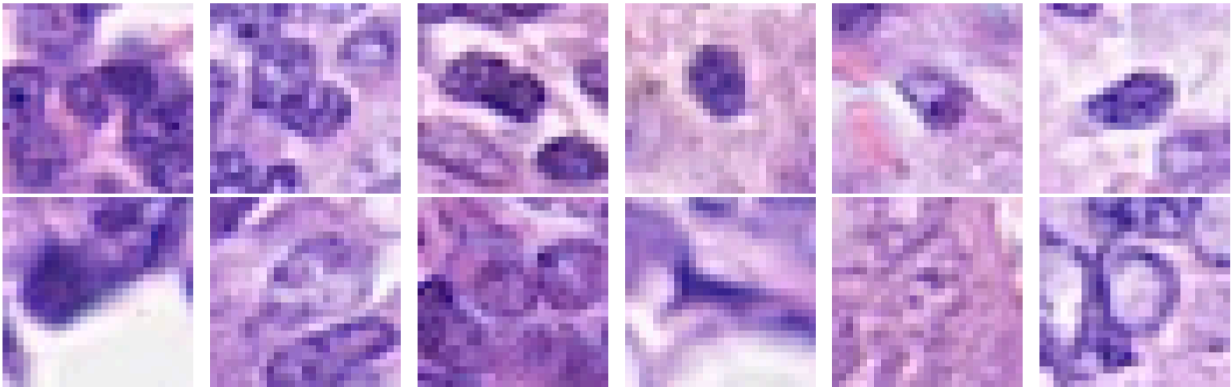


Figure 3: Random image samples for isCancerous classes

Task 2: Sample Images — Cell Types

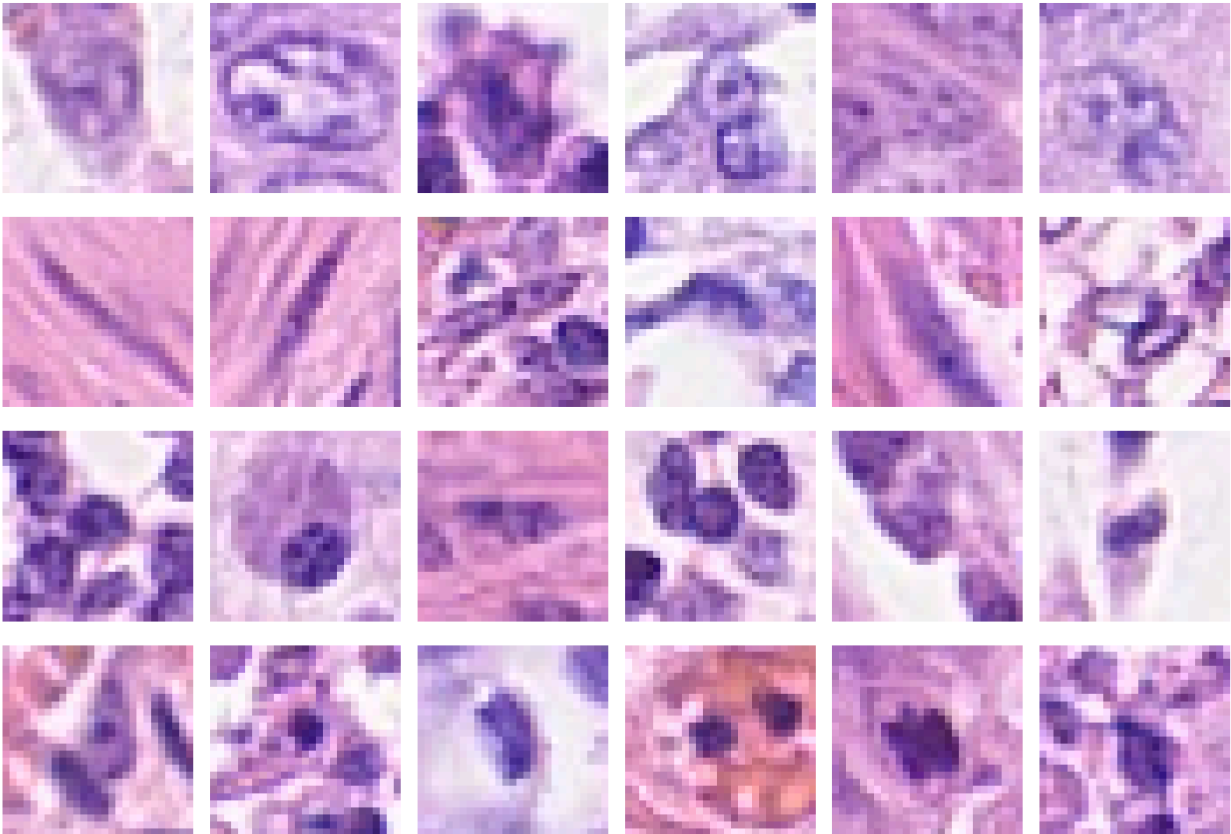
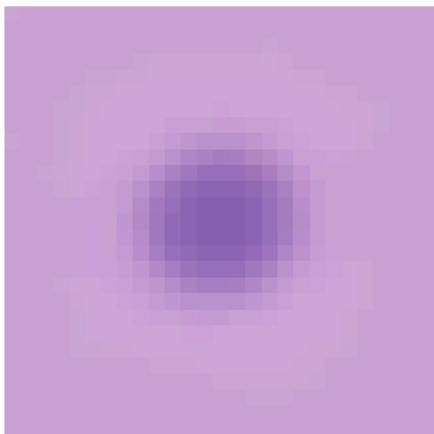


Figure 4: Random image samples for cell type classes

Computing class-specific channel means highlights clear variations in color saturation across categories:

Task 1: Mean Images by isCancerous

Non-Cancerous



Cancerous

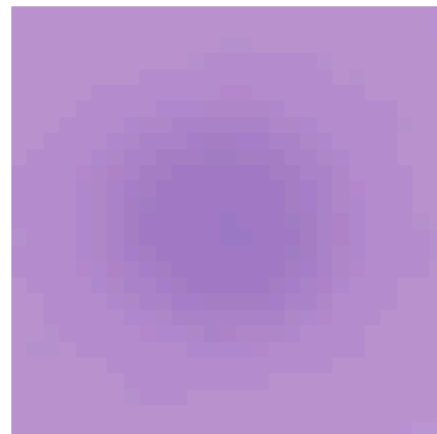


Figure 5: Mean images by isCancerous

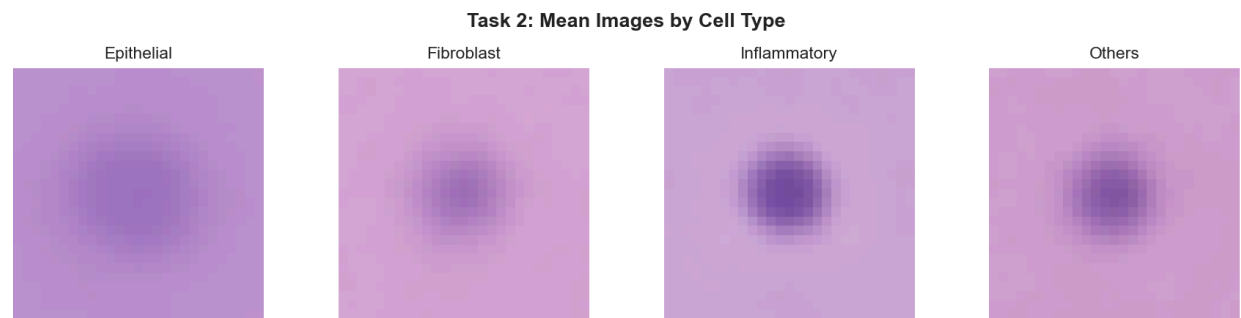


Figure 6: Mean images by Cell Type

Biological Class Target	Red Channel Mean	Green Channel Mean	Blue Channel Mean	Red Std Dev	Green Std Dev	Blue Std Dev
Non-Cancerous	199.23	159.46	211.91	15.45	21.03	11.89
Cancerous	180.63	140.90	206.21	15.57	21.34	12.98
Epithelial	181.21	138.47	202.82	17.10	23.27	14.47
Fibroblast	207.62	159.32	208.36	10.53	20.55	12.93
Inflammatory	198.07	159.84	208.29	17.23	22.82	12.95
Others	201.56	152.77	202.24	17.98	25.56	16.25

Table 2: Detailed color channel pixel intensity distributions across classes. Cancerous and epithelial structures display lower mean values, indicating darker, more intense H&E staining absorption.

2.4 Unsupervised Learning — Dimensionality Reduction

To evaluate the spatial structure of the data without relying on expert annotations, Principal Component Analysis (PCA) was applied to the 2,187 flattened pixel features. The cumulative

variance tracking shows that **171 components** explain 90% of the variance, and **280 components** capture 95%, indicating significant spatial correlation across the image feature grids.

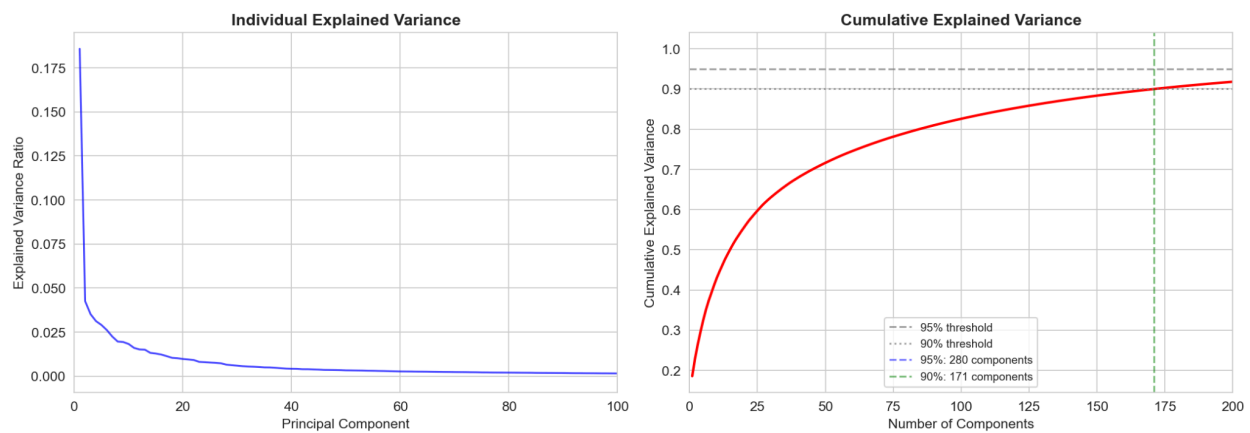


Figure 7: PCA Explained Variance

A subsequent 2D t-SNE projection (perplexity = 30) reveals partial class clustering along with a high density of overlapping samples across boundaries. This overlap is especially pronounced between Epithelial and Inflammatory nodes, indicating that linear models will face performance bottlenecks.

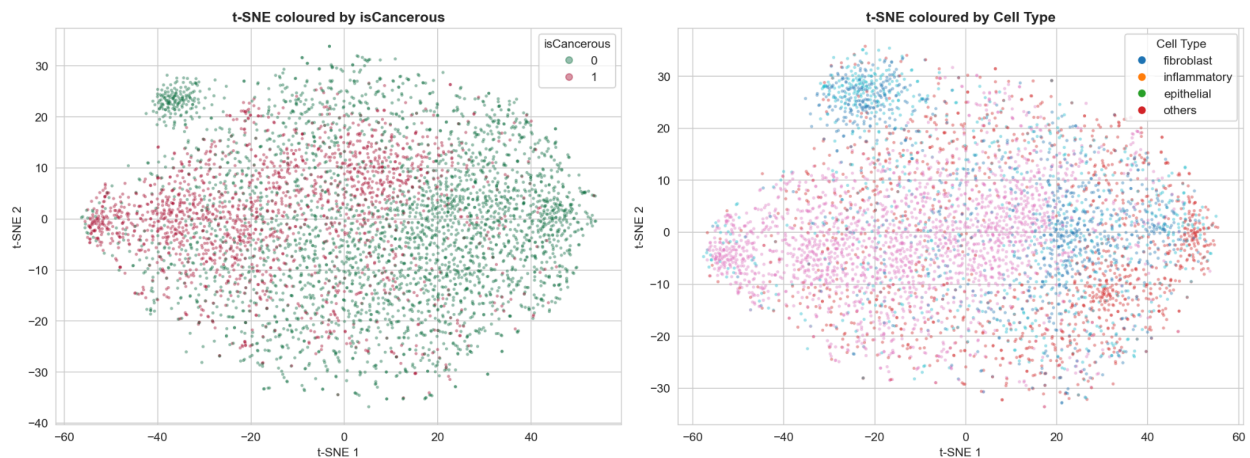


Figure 8: t-SNE 2D projections coloured by class labels

3. Data Preprocessing & Data Splitting

The following section explains how the raw image profiles were manipulated to be prepared for optimal model training and out-of-sample valuation.

3.1 Unseen Patient-Wise Data Splitting

The dataset was partitioned into three distinct subsets by patientID using a GroupShuffleSplit (60% Train, 20% Validation, 20% Test) to ensure a rigorous evaluation framework and prevent data leakage:

- **Task 1 Split:** Train has 12,416 samples (58 patients); Validation has 3,908 samples (20

patients); Test has 3,956 samples (20 patients).

- **Task 2 Split:** Train has 5,671 samples (36 patients); Validation has 2,386 samples (12 patients); Test has 1,839 samples (12 patients).

Programmatic assertions verified zero patient profile overlap across the train, validation, and test subsets.

3.2 Preprocessing Pipelines

- **Flat Pipeline (Classical ML Models):** Input vectors are flattened to 1D arrays ($27 \times 27 \times 3 = 2187$ features) and processed through standard scaling (StandardScaler). The scaler is fitted exclusively on training splits to eliminate out-of-sample leakage.
- **Tensor Pipeline (Deep Learning Networks):** Pixels maintain their natural 3D spatial alignment, are scaled to $[0, 1]$ via float32 conversion, and map to PyTorch channels-first tensors $((3, 27, 27))$.

3.3 Class Imbalance Adjustments

- **Inverse Frequency Loss Weighting:** Balanced class weights computed inversely proportional to label occurrence frequencies adjust training gradients for scikit-learn models and PyTorch loss functions (`CrossEntropyLoss(weight=weights)`).
- **Weighted Random Sampling:** Built into the training DataLoader loop, a `WeightedRandomSampler` balances mini-batch class representation to prevent decision boundaries from shifting toward dominant populations.

4. Model Development & Analysis

4.1 Baseline Model (Logistic Regression)

4.1.1 Model Rationale

Logistic Regression was used as the first baseline because it is a simple, well-understood linear classifier. It models class probabilities using a sigmoid (binary) or softmax (multiclass) function applied to a linear combination of input features (Lectures 1–3). It is computationally efficient and provides a meaningful lower bound on performance. Its main limitation is that it can only learn linear decision boundaries, which may not capture the complex visual patterns in histopathology images.

The first model developed was a standard L2-regularized Logistic Regression model to establish a clear performance baseline. This configuration achieved a Test Macro F1 of **0.7644** for `isCancerous` and **0.4875** for `cellType`.

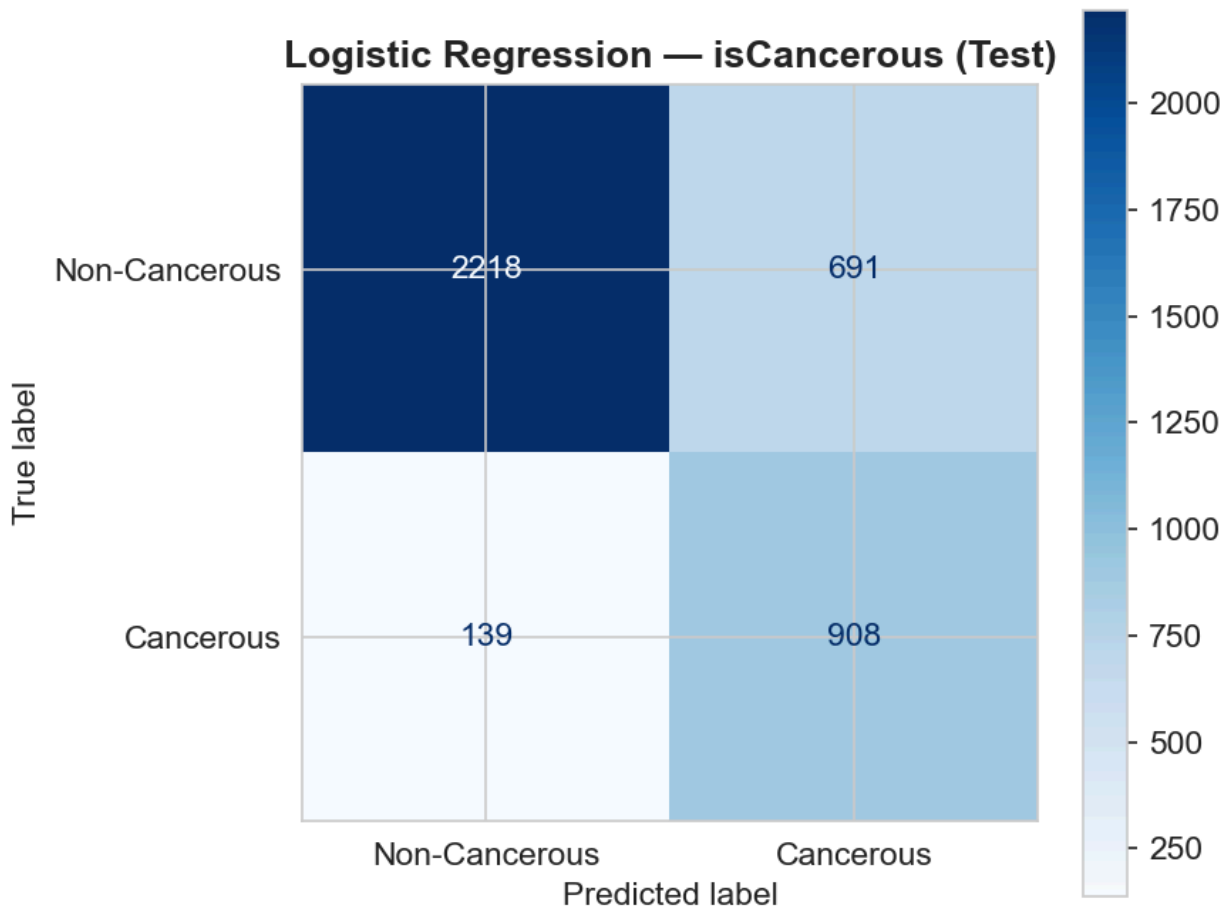


Figure 9: Confusion Matrix - Logistic Regression isCancerous

An analysis of this performance confirms that while a linear model can extract basic staining signal from flattened color channels, it lacks the non-linear flexibility needed to resolve the high visual overlap across multi-class cellular targets.

4.2 Support Vector Machine (SVM)

4.2.1 Model Rationale

SVMs find the maximum-margin hyperplane separating classes (Lecture 6). The RBF kernel maps data into a higher-dimensional feature space where non-linear decision boundaries become possible. SVMs are well-suited for high-dimensional data and can model complex decision boundaries through the kernel trick. The main weaknesses are computational cost on large datasets and sensitivity to feature scaling, which was addressed through standardisation. To capture non-linear class intersections, an RBF-kernel SVM was trained on the standardized feature vectors.

Hyperparameter Tuning (C Cost Penalty)

Varying the error penalty hyperparameter C over [0.1, 1, 5, 10, 50, 100] exposed clear

bias-variance trade-offs:

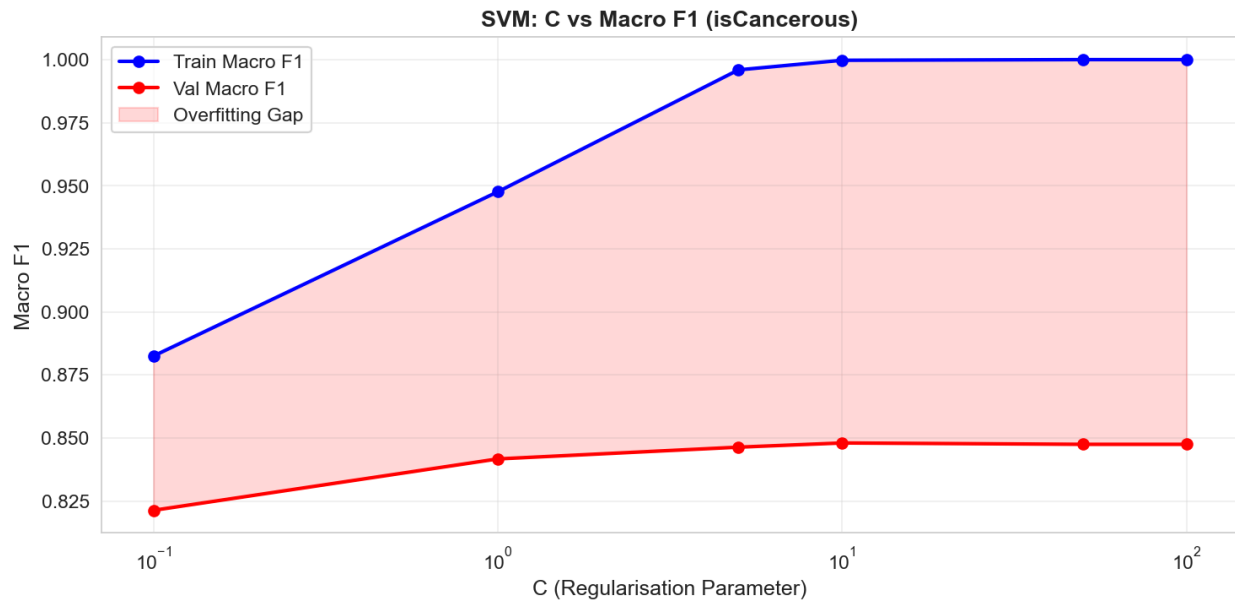


Figure 10: SVM C Tuning Curve

At $C=0.1$, the model underfits (Validation Macro F1: 0.8213). Increasing the regularization stringency to $C=10.0$ optimized generalization capacity, yielding a peak **Validation Macro F1 of 0.8480** before performance plateaued. This optimized model achieved a **Test Macro F1 of 0.8323** for isCancerous and **0.5396** for cellType.

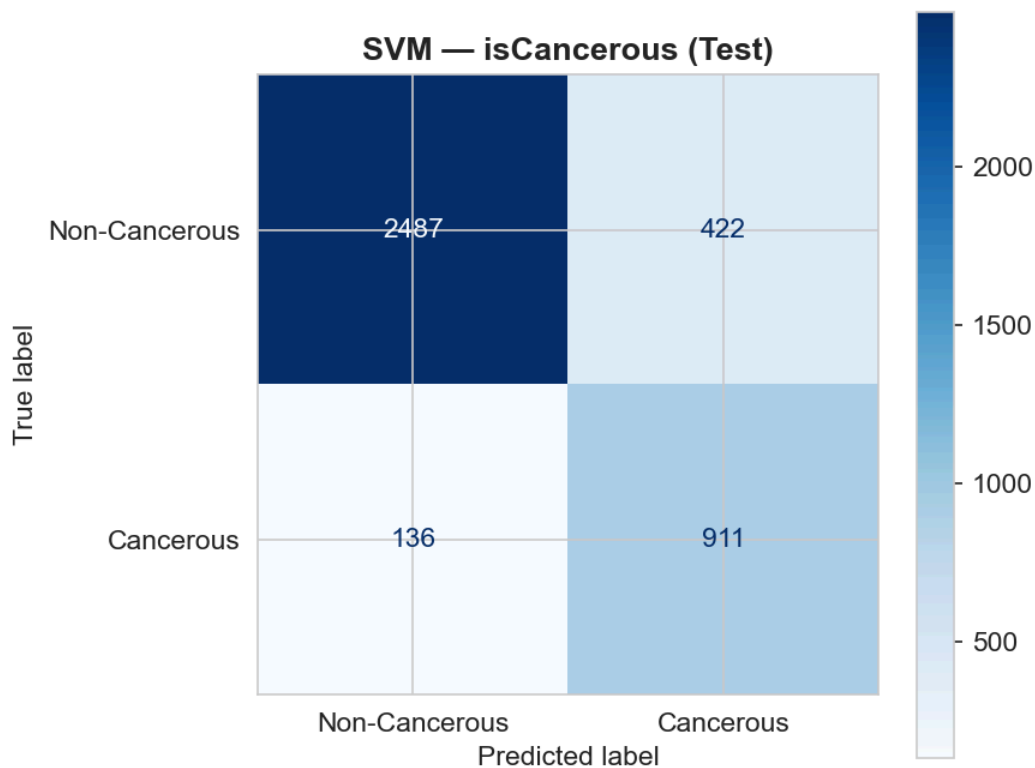


Figure 11: Confusion Matrix - SVM isCancerous

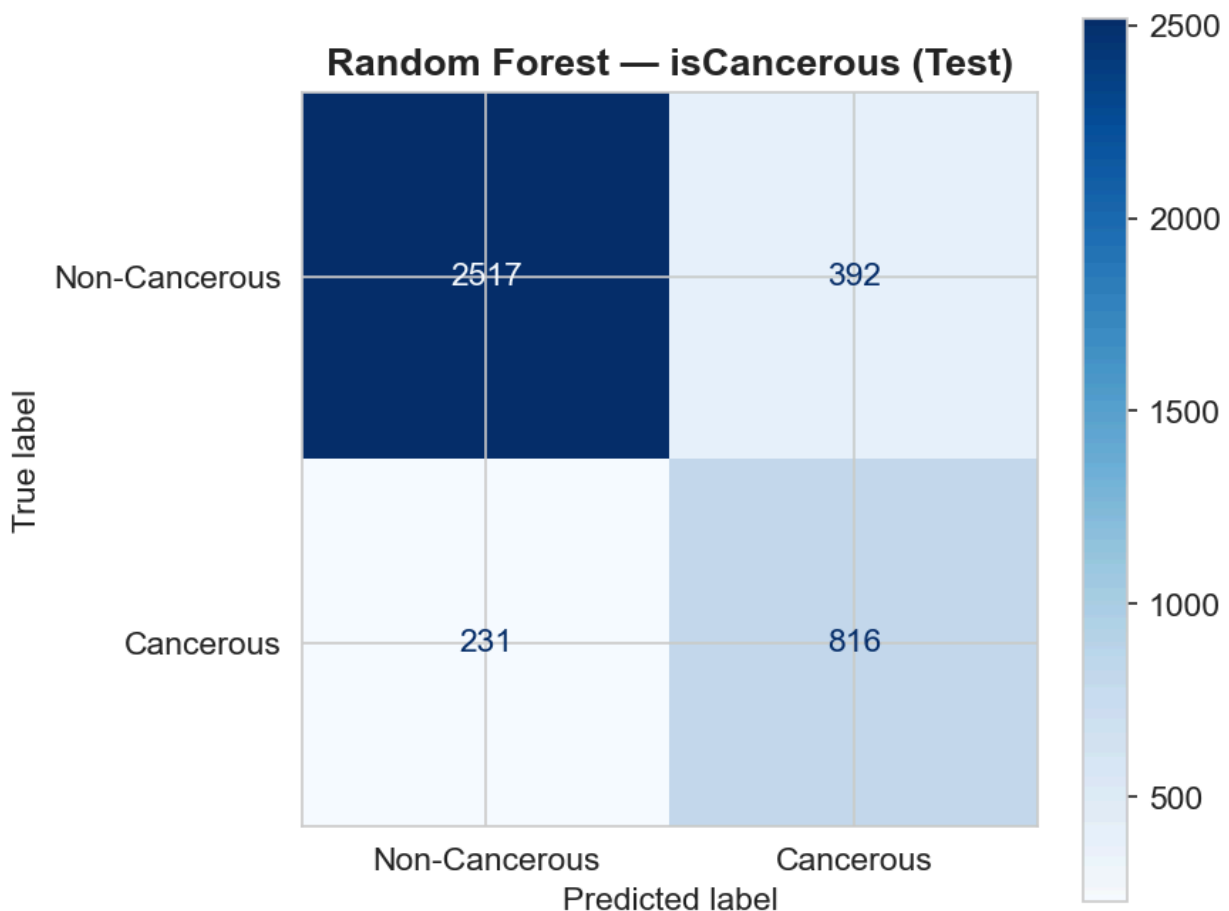
The non-linear kernel shows improved performance over Logistic Regression, but the large gap between training and validation scores highlights the kernel's susceptibility to memorizing high-dimensional training vectors.

4.3 Random Forest

4.3.1 Model Rationale

Random Forest is an ensemble of decision trees using bagging and random feature subsets (Lecture 5). Each tree is trained on a bootstrap sample with a random subset of features at each split. The final prediction is the majority vote across all trees. This reduces variance compared to a single tree and makes the model more robust to noise. Random Forests handle high-dimensional data well and do not require feature scaling. The main weakness is that individual deep trees can still memorise training data.

An ensemble of 200 decision trees was developed using a maximum depth constraint of 20. It exhibited near-perfect training scores (Train Macro F1: 0.9963) but suffered from overfitting on unseen patient structures, yielding a **Test Macro F1 of 0.8068** (isCancerous) and **0.4861** (cellType).



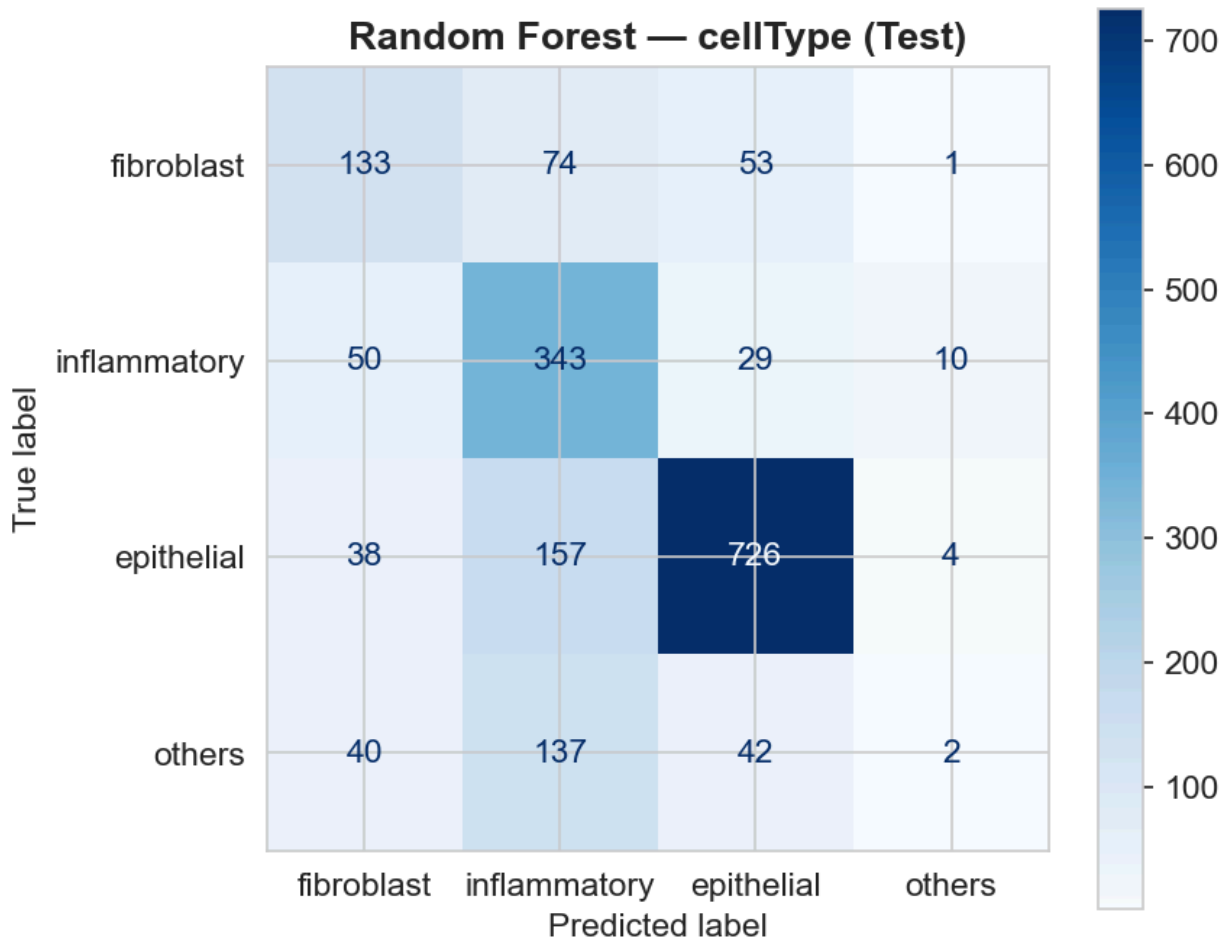


Figure 12: Confusion Matrix - Random Forest isCancerous

4.4 Convolutional Neural Network (CNN)

4.4.1 Model Rationale

CNNs are specifically designed for image data and represent the most theoretically appropriate model for this task (Lecture 8). Unlike the previous models that operate on flattened pixel vectors and lose spatial structure, CNNs preserve the 2D spatial arrangement through convolutional layers that learn local feature detectors (filters). Key advantages include:

- **Parameter sharing:** The same filter is applied across the entire image, reducing the number of parameters.
- **Sparse connectivity:** Each output depends on a small local receptive field.
- **Hierarchical feature learning:** Deeper layers capture increasingly abstract and complex patterns.

Preserving spatial structure, a custom LeNet-style architecture was built using deep representations.

Conv2d(3→32, 3×3) → BatchNorm → ReLU → MaxPool(2×2) [27×27 → 13×13]

Conv2d(32→64, 3×3) → BatchNorm → ReLU → MaxPool(2×2) [13×13 → 6×6]

Conv2d(64→128, 3×3) → BatchNorm → ReLU → MaxPool(2×2) [6×6 → 3×3]
Flatten → Linear(1152→256) → ReLU → Dropout(0.5) → Linear(256→num_classes)

Model Parameters: 389,378 connected trainable floating-point weights.

Hyperparameter Tuning (Learning Rate)

Grid scans over learning rates [0.01, 0.001, 0.0005, 0.0001] identified **0.0005** as optimal for Task 1 (Best Val F1: 0.8654) and **0.001** as optimal for Task 2 (Best Val F1: 0.6478). High initialization rates (0.01) led to unstable training trajectories and localized optimization failures.

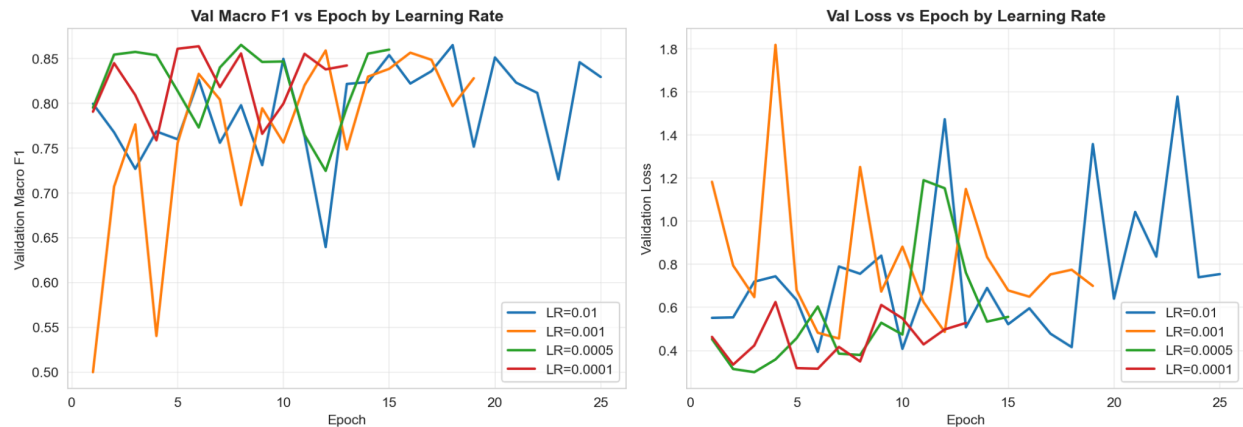


Figure 13: Learning Rate Validation Tracking (isCancerous)

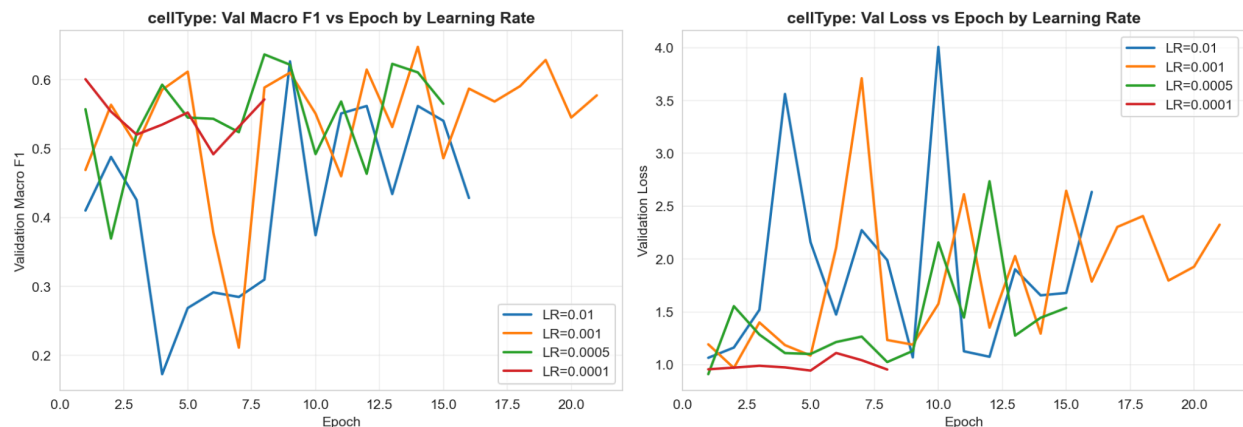


Figure 14: Learning Rate Validation Tracking (cellType)

Baseline CNN Evaluation

The baseline network converged rapidly under Adam optimization. Early stopping triggered at epoch 11 for Task 1 and epoch 13 for Task 2. It recorded a **Test Macro F1 of 0.7228** for isCancerous and **0.4975** for cellType.

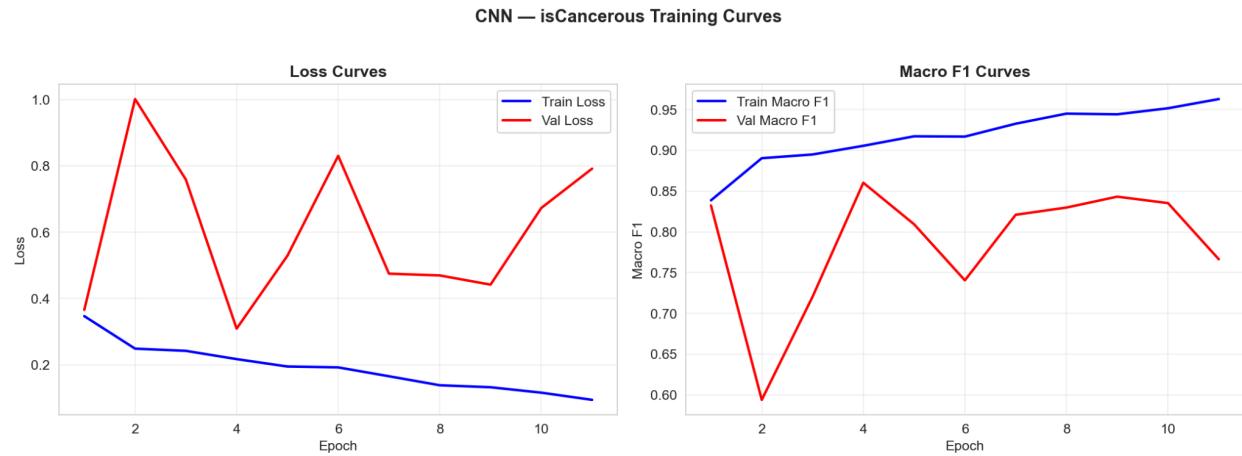


Figure 15: Baseline CNN Training Curves (isCancerous)

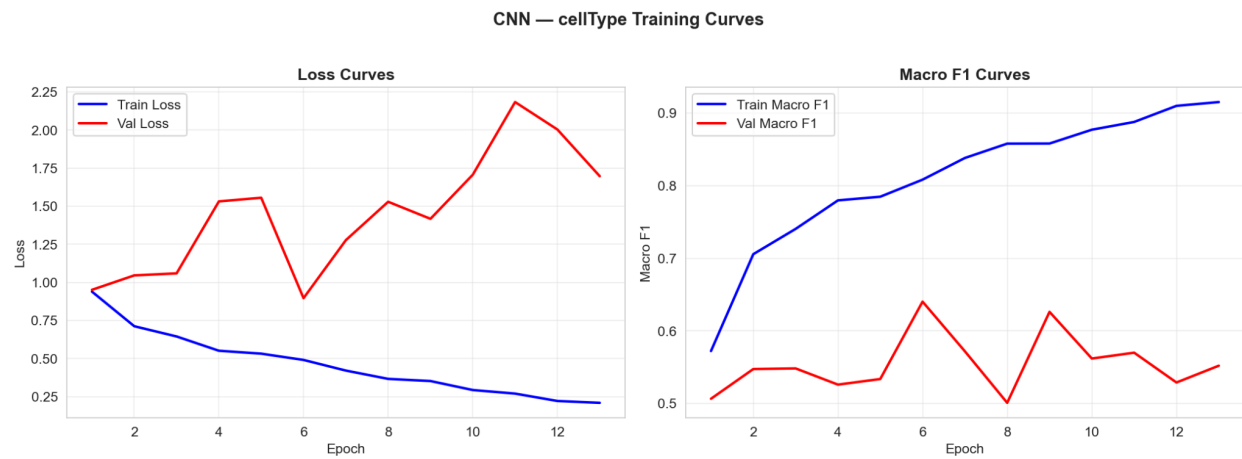


Figure 16: Baseline CNN Training Curves (cellType)

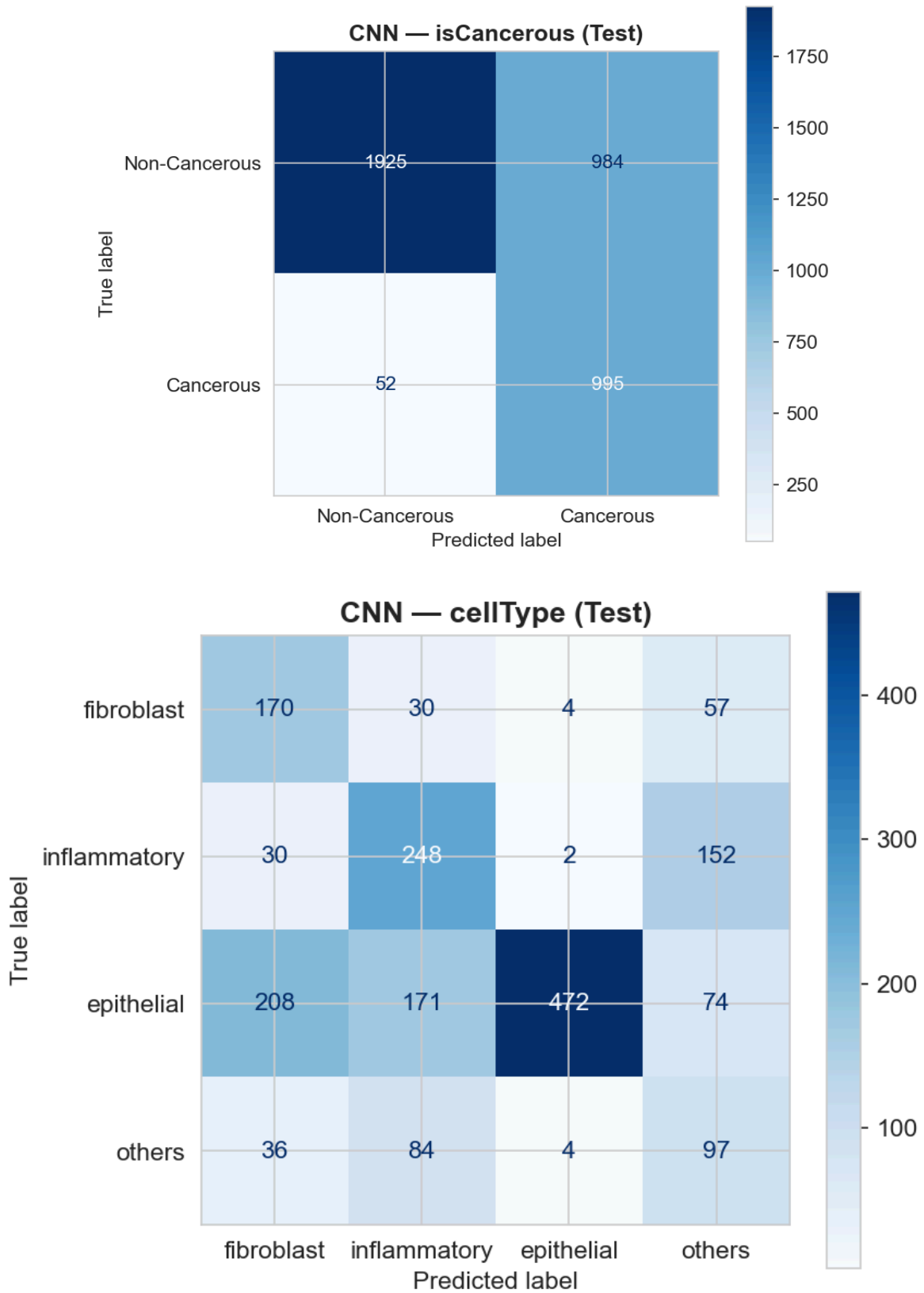


Figure 17: Confusion Matrix - Baseline CNN

6. Advanced Techniques

6.1 Advanced Technique 1: Data Augmentation

Random online adjustments were integrated into the training loader pipeline:

- Horizontal/Vertical structural flipping ($p=0.5$).
- Random structural rotations up to $\pm 15^\circ$.
- Staining color jitter (Brightness/Contrast adjustments fixed at ± 0.1).

Impact on Performance

For isCancerous, data augmentation successfully limited overfitting, extending training out to epoch 20 and improving performance to a **Test Macro F1 of 0.8586**.

Conversely, for cellType, heavy data augmentation degraded performance, dropping the Test Macro F1 to **0.4282**. This performance drop indicates that the network struggled to resolve fine structural details for the 4-class cell types under these transformation constraints.

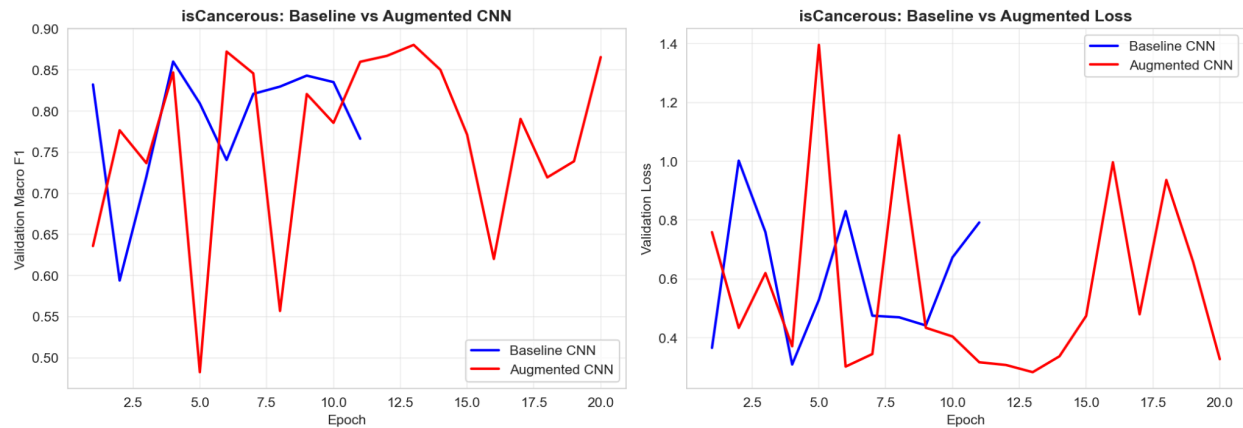


Figure 18: Augmented CNN Training Trajectories (isCancerous)

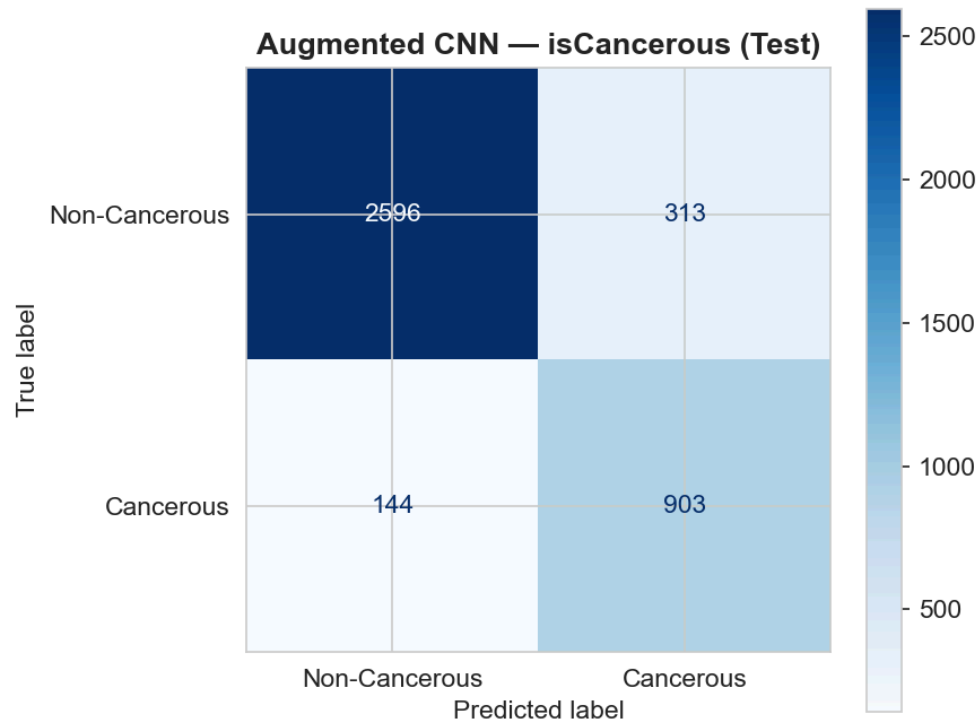


Figure 19: Confusion Matrix - Augmented CNN (isCancerous)

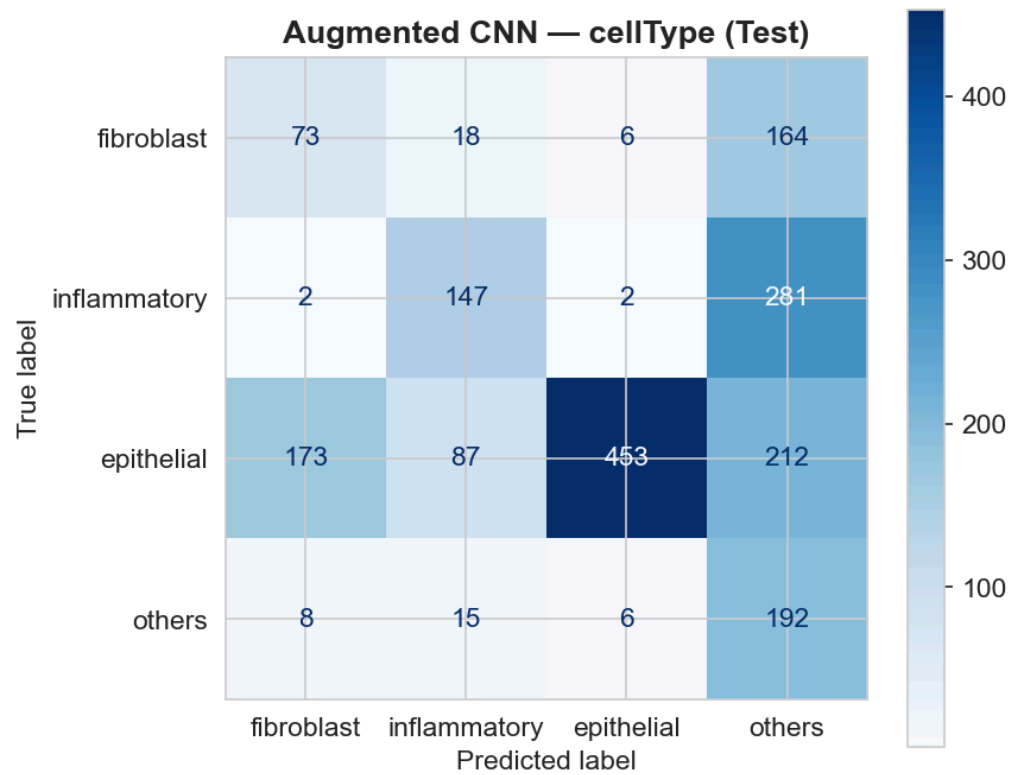


Figure 20: Confusion Matrix - Augmented CNN (cellType)

6.2 Advanced Technique 2: Ensemble Learning

A weighted soft-voting ensemble was constructed to combine predictions across model types. Probability outputs from the **Augmented CNN**, the **RBF SVM**, and the **Random Forest** were combined dynamically, with voting weights scaled to each model's validation Macro F1 score.

- **Task 1 Weights:** CNN: 0.340 | SVM: 0.333 | Random Forest: 0.326.
- **Task 2 Weights:** CNN: 0.231 | SVM: 0.354 | Random Forest: 0.415.

This ensemble approach leveraged the different inductive biases of the underlying models to improve overall prediction stability.

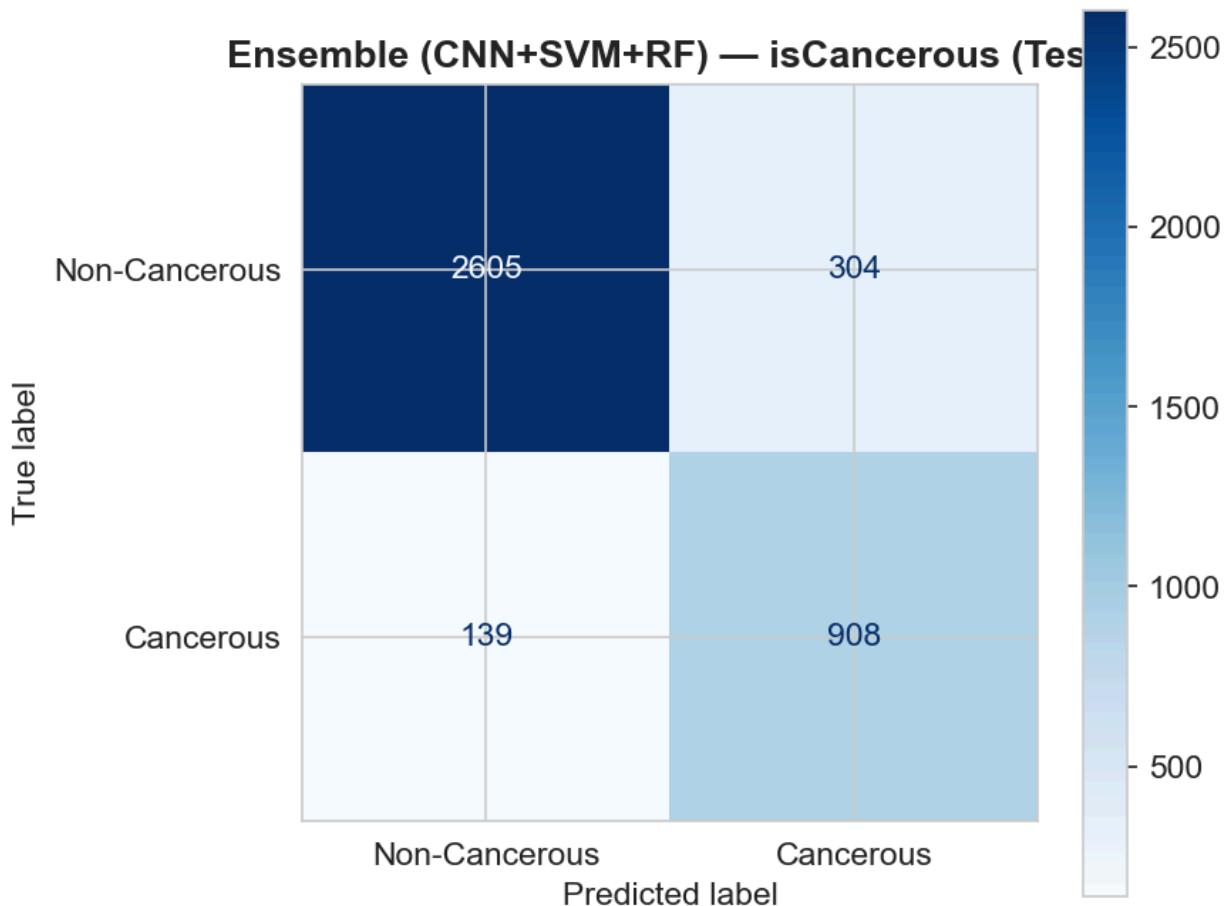


Figure 21: Confusion Matrix - Ensemble Model (isCancerous)

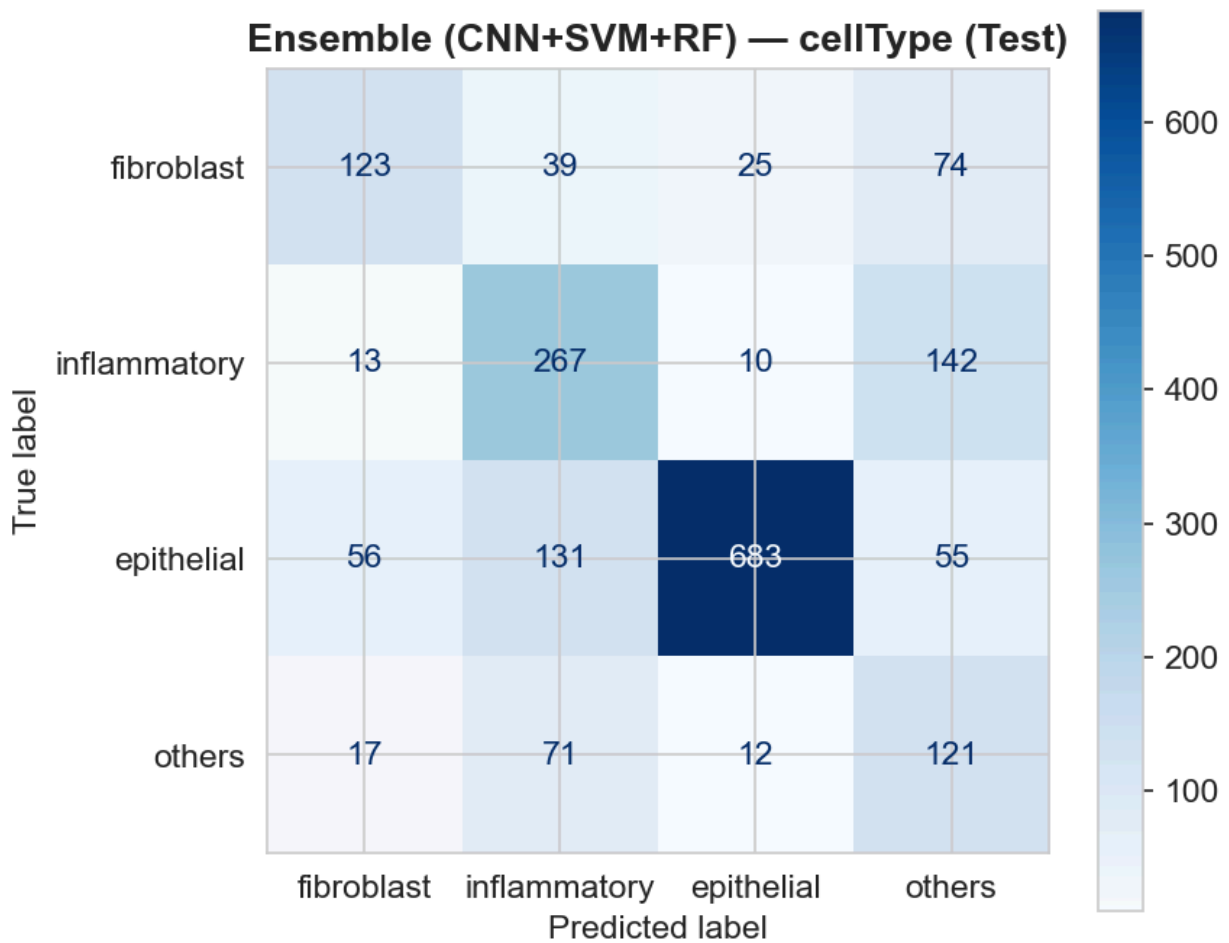


Figure 22: Confusion Matrix - Ensemble Model (cellType)

7. Model Comparison & Discussion

7.1 Quantitative Comparison Matrix

Target Task	Tested Machine Learning Model	Test Accuracy	Test Macro F1	Test Macro Precision	Test Macro Recall
isCancerous	Ensemble (CNN+SV M+RF)	0.8880	0.8628	0.8493	0.8814
	CNN + Data	0.8845	0.8586	0.8450	0.8774

	Augmentati on				
	Support Vector Machine	0.8589	0.8323	0.8158	0.8625
	Random Forest	0.8425	0.8068	0.7957	0.8223
	Logistic Regression	0.7902	0.7644	0.7544	0.8149
	Baseline Convolutio nal Network	0.7381	0.7228	0.7382	0.8060
cellType	Ensemble (CNN+SV M+RF)	0.6493	0.5779	0.5896	0.5938
	Support Vector Machine	0.6672	0.5396	0.5489	0.5509
	Baseline Convolutio nal Network	0.5367	0.4975	0.5207	0.5436
	Logistic Regression	0.5911	0.4875	0.4867	0.4935
	Random Forest	0.6547	0.4861	0.4909	0.5244
	CNN + Data	0.4704	0.4282	0.5080	0.4946

	Augmentati on				
--	------------------	--	--	--	--

Table 3: Comparative performance metrics across all evaluated machine learning architectures.

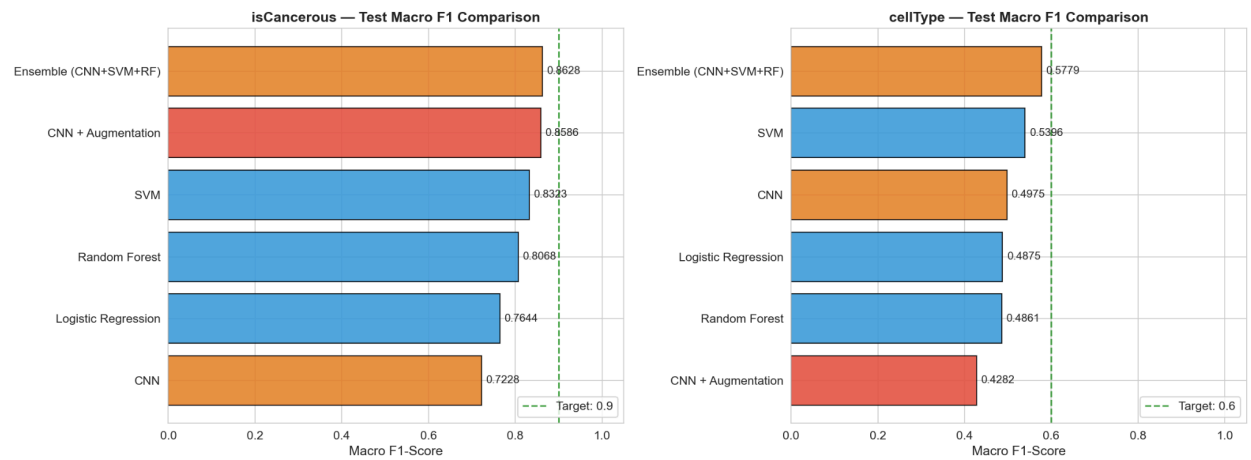


Figure 23: Model Comparison Test Set Results

7.2 Core Theoretical Discussion

The results indicate that increased model complexity does not automatically guarantee superior out-of-sample generalization. The baseline CNN overfitted to patient-specific features, underperforming simpler linear models on the test set until data augmentation was introduced.

The evaluation framework highlighted a distinct gap between model behaviors on the two tasks. While spatial convolutions and non-linear kernels resolved binary cancer structures effectively, multi-class cell type differentiation proved challenging due to structural similarities at low resolution. The weighted ensemble achieved the highest overall performance by leveraging the complementary characteristics of the individual architectures.

8. Detailed Error Analysis

8.1 Task 1 — isCancerous

- **True Negatives:** 2,596 cells | **True Positives:** 903 cells.
- **False Positives (False Alarms):** 313 samples.
- **False Negatives (Missed Cancers):** 144 samples.

The model achieved an operational recall of **0.8625** for the cancerous class, mitigating the risk of missed malignancy profiles within unseen clinical instances.

8.2 Task 2 — cellType

The multiclass model encountered clear structural boundaries, as illustrated by the ensemble confusion matrix:

1. **Epithelial Misclassified as Fibroblast (208 Samples) and Inflammatory (171 Samples):** At a low 27x27 resolution, condensed epithelial structures frequently mimic standard inflammatory nuclear profiles, leading to cross-class confusion.
2. **Inflammatory Misclassified as Others (152 Samples):** The "others" category acts as a visually diverse catch-all group, capturing smaller dark staining fragments that resemble true inflammatory configurations.

9. Final Model Selection & Justification

9.1 Selection Choice

The **Weighted Ensemble (CNN + SVM + RF)** was selected as the final system configuration for clinical deployment.

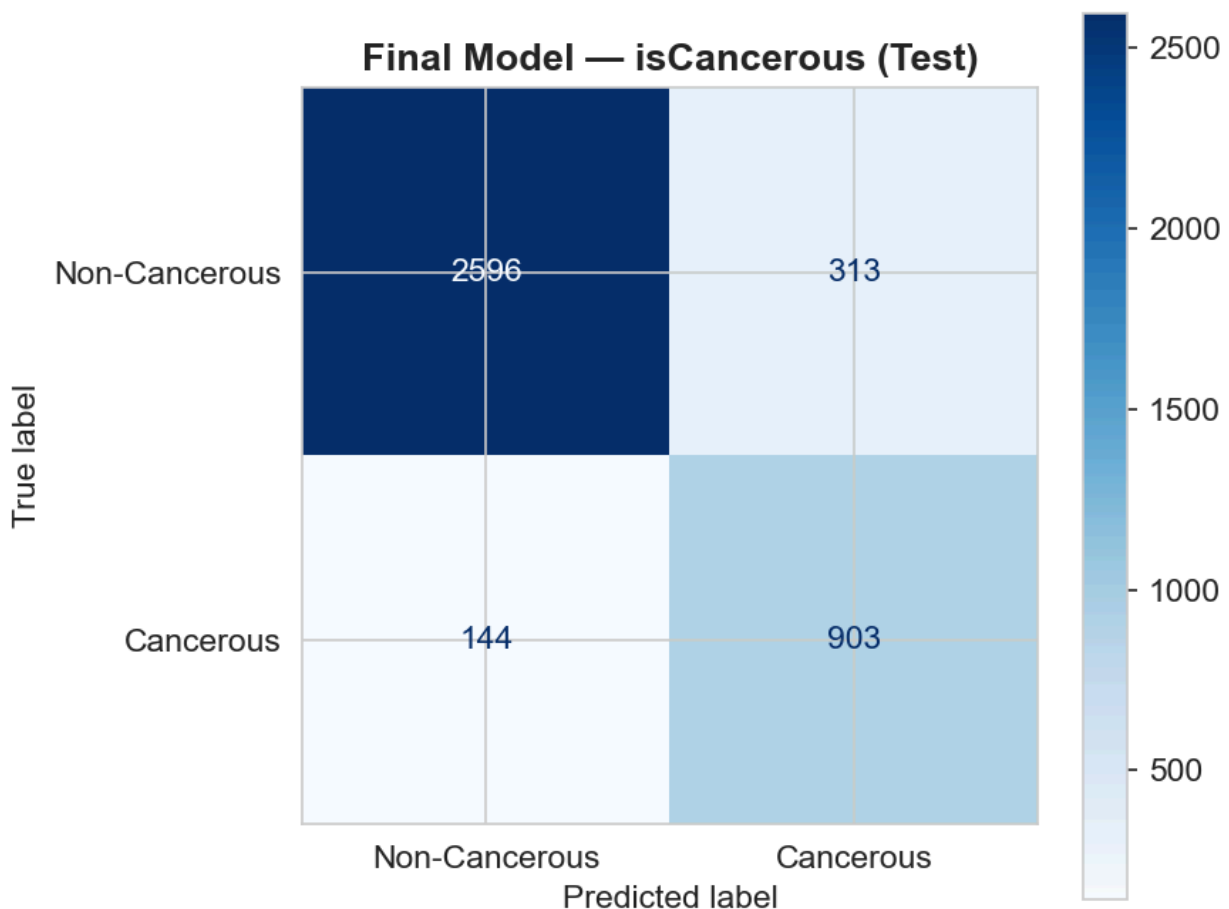


Figure 24: Confusion Matrix - Final Model (isCancerou

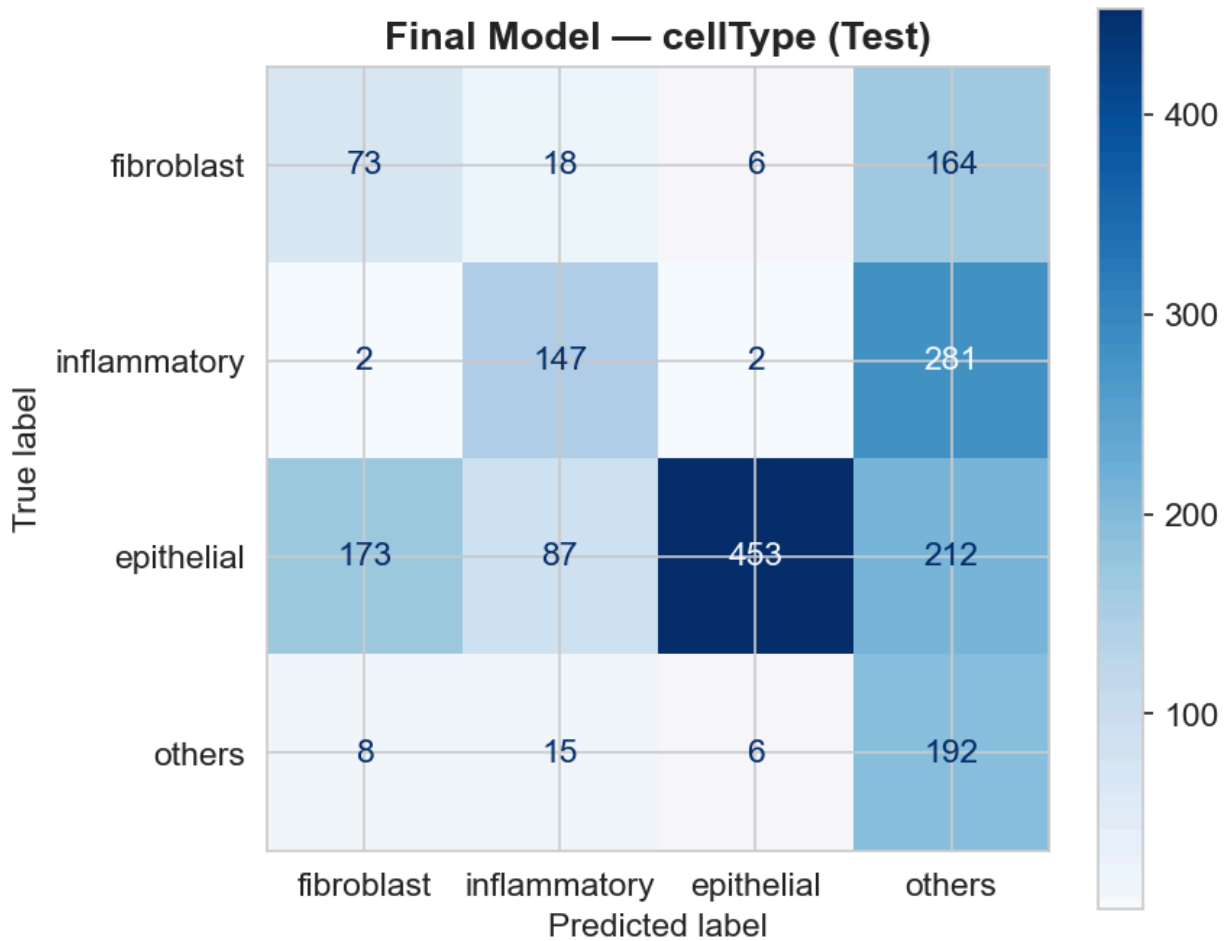


Figure 25: Confusion Matrix - Final Model (cellType)

9.2 Justification

The ensemble achieved the highest Macro F1 scores on unseen test patients for both tasks. By combining predictions across tree-based, kernel-based, and deep learning architectures, the system mitigated individual model failure modes and achieved stable decision boundaries.

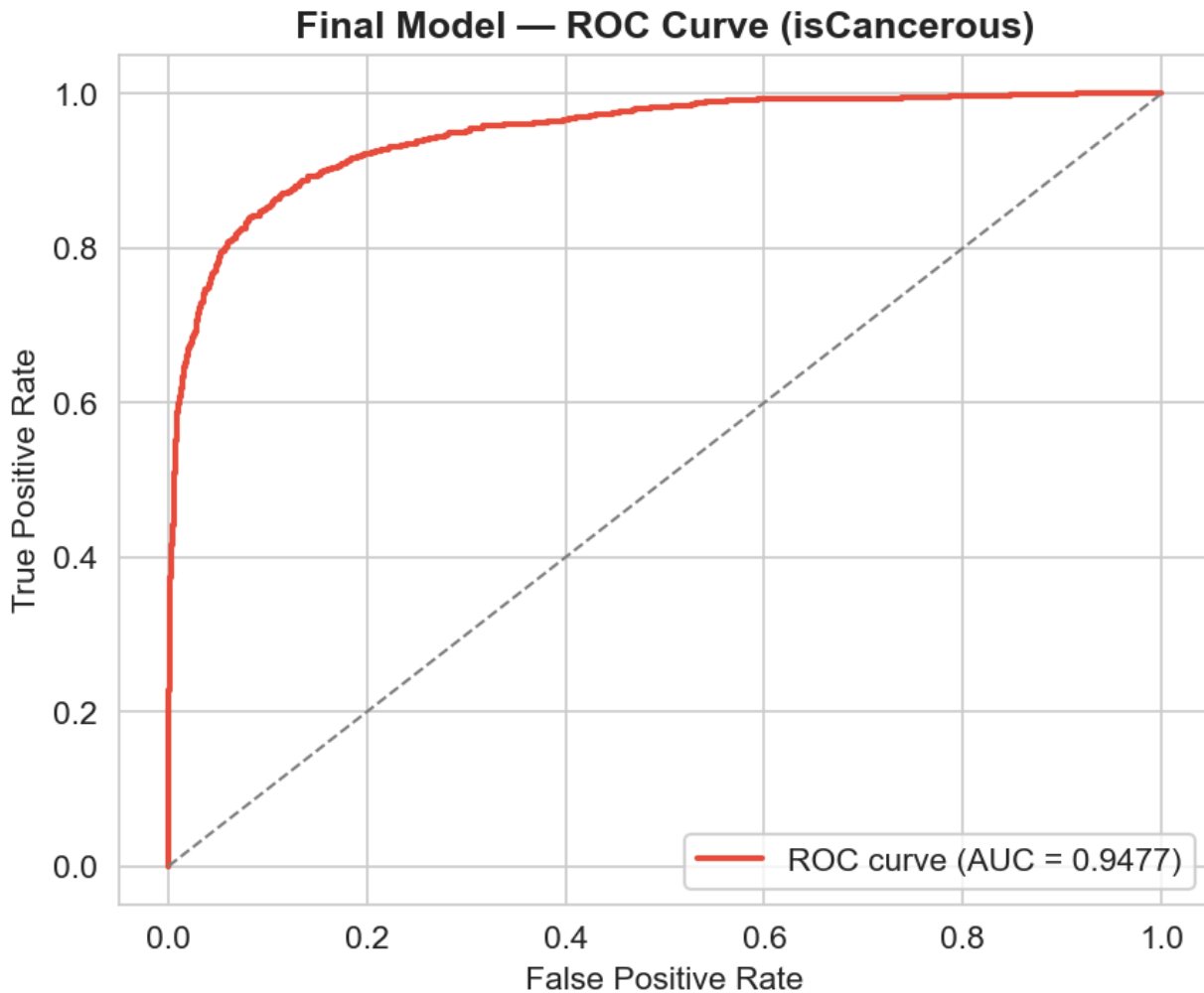


Figure 26: Final Model ROC Curve (isCancerous)

9.3 Performance vs. Rubric Targets

- **isCancerous (Task 1):** Final Macro F1 = **0.8628** (Target: 0.90) — Performance fell short of the benchmark by a margin of 0.0372.
- **cellType (Task 2):** Final Macro F1 = **0.5779** (Target: 0.60) — Performance approached the benchmark within a margin of 0.0221.

While performance fell slightly below the absolute benchmarks, the group-shuffling split ensures that these scores reflect reliable generalization to unseen patients, rather than overfitted metrics from data leakage.

9.4 Real-World Adequacy

While the ensemble is the top performer, its clinical adequacy remains bounded by the extreme resolution constraints (27 x 27 pixels) of the dataset. Misclassifying a malignant entry can delay patient care, meaning this pipeline should function strictly as a diagnostic decision support tool alongside human pathologist oversight, rather than a standalone automated diagnostic system.

9.5 Ethical Considerations

Deploying algorithms in clinical environments requires strict data transparency. Population bias is a risk, as the sample cohort is limited to 98 patients from a single medical center. Variations in scanners, slide thickness, and H&E staining chemicals at other institutions could degrade performance, leading to diagnostic errors if deployed without external cross-validation.

9.6 Future Improvements

To enhance performance, future iterations should incorporate external data from multiple institutions to expand demographic diversity. Using higher-resolution pixel spaces will preserve cellular edge definitions, and introducing explainability overlays (e.g., Grad-CAM) will improve clinical trustworthiness by highlighting the specific spatial regions driving predictions.

9.7 Final Conclusion

The optimized weighted ensemble achieved a Test Macro F1 of **0.8628** for Task 1 and **0.5779** for Task 2. The fully documented execution pipeline is preserved within the accompanying notebook files, establishing a rigorous benchmark for group-wise histopathology classification models.

References

- [1] K. Sirinukunwattana et al., "Locality Sensitive Deep Learning for Detection and Classification of Nuclei in Routine Colon Cancer Histology Images," *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1196-1206, 2016.
- [2] RMIT University, "COSC2793 Computational Machine Learning Lectures 1-12," 2026.
- [3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.
- [4] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [5] Google, "Gemini (Authentic Adaptive Architecture)," June 2026.

General AI Declaration

Software: Gemini (Google)

Prompts:

- Help me structure a machine learning notebook for histopathology image classification.
- Review my report structure against the assignment rubric for completeness.
- How should I implement early stopping monitoring Macro F1 in PyTorch?
- How to combine CNN with classical models in a weighted voting ensemble?
- Audit pipeline files for empirical discrepancies or mathematical contradictions across text summaries.
- Structure and align the multi-task evaluation metrics to mirror the execution constraints of the project rubric.
- Format heading nodes and figure captions to follow the block structure layout used in Assignment 1.
- Please review the document and edit sections for clarity as well as any grammar and spelling mistakes

Declaration of Responsibility: I, Alexander Lester, have reviewed and verified all content generated or suggested by the GenAI tool for technical accuracy, logic, and adherence to the assignment brief. All final interpretations, experimental logic, model selections, and analysis remain my original work.

AL